



DZPHARM • PLATFORM TRANSFORMATION

# Sprint Planning

The week-by-week execution plan for the 13-week platform transformation. Every task has a ticket ID, title, assignee, effort estimate, dependencies, definition-of-done, acceptance criteria, and risk flag. Jira-ready — copy the ticket IDs directly into your backlog.

**13**

SPRINTS

**~80**

TICKETS

**4**

TEAM MEMBERS

**~13w**

TOTAL DURATION

SOURCE

# Sprint Structure & Team

The 13-week transformation is organized into 13 one-week sprints. Each sprint has 2-8 tickets across 4 parallel tracks. The team is 2 engineers + 1 designer + 1 PM. Ticket IDs follow the pattern `[PILLAR]-[SPRINT]-[SEQ]` (e.g., `R-01-01` = Routing, Sprint 1, Ticket 1).

## Team Roles

| ROLE                    | PERSON  | TRACK        | RESPONSIBILITY                                                            |
|-------------------------|---------|--------------|---------------------------------------------------------------------------|
| Engineer A (Senior)     | Track A | FOUNDATION   | Routing (#1), State (#2), Auth (#8) — the three P0 pillars                |
| Engineer B (Mid-Senior) | Track B | DESIGN       | Typography (#54), Color System (#53), Copilot (#36) — the three upgrades  |
| Designer                | Track C | VERIFICATION | Palette verification, Storybook, color-blindness testing, visual QA       |
| Product Manager         | Track D | COORDINATION | Feature flags, compliance docs, rollout management, stakeholder reporting |

## Ticket ID Prefixes

| PREFIX                | PILLAR                      | OWNER TRACK |
|-----------------------|-----------------------------|-------------|
| <code>R-XX-YY</code>  | #1 Routing Architecture     | Track A     |
| <code>A-XX-YY</code>  | #8 Authentication Model     | Track A     |
| <code>S-XX-YY</code>  | #2 State Management         | Track A     |
| <code>C-XX-YY</code>  | #53 Color System            | Track B + C |
| <code>T-XX-YY</code>  | #54 Typography              | Track B     |
| <code>P-XX-YY</code>  | #36 Copilot Core            | Track B     |
| <code>D-XX-YY</code>  | Designer / Verification     | Track C     |
| <code>PM-XX-YY</code> | PM / Coordination           | Track D     |
| <code>X-XX-YY</code>  | Cross-cutting / Independent | Any         |

## Sprint Cadence

Each sprint is 1 calendar week (Monday → Friday). Monday: sprint planning (30 min). Tuesday-Thursday: implementation. Friday: demo + retrospective (1 hour). Feature flags are toggled on Wednesdays (mid-

sprint checkpoint). The PM monitors error rates daily during rollout phases.

## Sprint 0 · Day 1 Quick Wins (1 Hour)

3 independent tickets · No dependencies · Ship before any pillar work begins

### T-00-01 Load Inter + Noto Sans Arabic + JetBrains Mono via next/font/google

Engineer B · 30 min

**DoD:** `document.fonts.size === 3` in browser console. Arabic text renders in Noto Sans Arabic (not Segoe UI). `getComputedStyle(body).fontFamily` contains `'__Inter'`.

**Depends on:** None

LOW

### X-00-02 Add prefers-reduced-motion CSS media query

Engineer A · 5 min

**DoD:** Enable 'Reduce motion' in macOS System Settings → Accessibility → Display. All animations (modal, toast, nav hover) are instant (0.01ms). No visible motion.

**Depends on:** None

LOW

### X-00-03 Add aria-labels to 4 icon-only buttons

Engineer B · 25 min

**DoD:** Screen reader announces 'Mode botanique, bouton', 'Nuancier — changer de palette, bouton', 'Ouvrir le Copilote IA, bouton', 'Passer en mode clair, bouton' (state-aware). NVDA/VoiceOver verification.

**Depends on:** None

LOW

## DAY 1 OUTCOME

1 hour of work. 3 features go from C/C+ to A (#54 Typography B → A-, #67 Motion C+ → A, #62 Icons C → A). Brand identity becomes stable across all devices. WCAG AAA for reduced motion. Ship in a single PR before any pillar work.

## Sprint 1 · Week 1 · Routing Phase 1 + Color Verification

Track A: #1 Routing Phase 1 (4 primary routes) · Track B+C: #53 Color Phase 1 (verification) · Track D: Feature flags + redirects

### R-01-01 Create app/repertoire/page.tsx with SSR data fetching + metadata

Engineer A · 4h

**DoD:** Direct URL `/repertoire` loads the Répertoire (no 404). Search params (`q`, `status`, `domain`, `page`) read from URL. Metadata title: 'Répertoire officiel des 9 555 médicaments AMM · DzPharm'. `loading.tsx` skeleton shows during fetch.

**Depends on:** None

### R-01-02 Create app/prix-chifa/page.tsx

Engineer A · 2h

**DoD:** Direct URL `/prix-chifa` loads the pricing view. Metadata present. `loading.tsx` skeleton.

**Depends on:** R-01-01 (pattern reuse)

#### R-01-03 Create app/interactions/page.tsx with URL state hook

Engineer A · 4h

**DoD:** Direct URL /interactions loads. Drug list in URL params (?drugs=8421,1523). Patient context in URL (?ctx=...). Refresh preserves drug list. URL is shareable.

**Depends on:** R-01-01

#### R-01-04 Create app/copilote/page.tsx (full-page view, not modal)

Engineer A · 2h

**DoD:** Direct URL /copilote loads the Copilot as a full-page view (not modal). Mode toggle works. Suggested prompts visible.

**Depends on:** R-01-01

#### R-01-05 Update header to use next/link instead of setState navigation

Engineer A · 4h

**DoD:** Header nav uses Répertoire. Active state via usePathname(). No more setActiveView() calls. Browser back/forward works between all 4 new routes.

**Depends on:** R-01-01 through R-01-04

#### D-01-01 Run all 20 Nuancier palettes through WCAG contrast checker (both themes)

Designer · 4h

**DoD:** Contrast report: each palette × 2 themes = 40 tests. Document which pass AA, which fail. Fix failures by adjusting palette colors (likely 3-5 palettes need light-theme variants).

**Depends on:** None

#### D-01-02 Run all 20 palettes through color-blindness simulator

Designer · 4h

**DoD:** Color-blindness report: 20 palettes × 3 simulators (deuteranopia, protanopia, tritanopia). Document which are safe. Fix any that fail.

**Depends on:** D-01-01

#### PM-01-01 Set up feature flag infrastructure (Vercel Edge Config or env vars)

PM · 2h

**DoD:** Feature flag 'new-routing-phase1' created. Default: off. Can toggle per-user or per-percentage. Rollout plan documented: 0% → 10% → 50% → 100%.

**Depends on:** None

#### PM-01-02 Create backward-compatibility redirects in next.config.js

PM · 1h

**DoD:** Old URLs (/?view=repertoire) redirect to /repertoire (permanent). 4 redirects configured. Tested with curl.

**Depends on:** R-01-01

## Sprint 2 · Week 2 · Routing Phase 2 + Sitemap

Track A: #1 Routing Phase 2 (7 clinical tools) · Track B: #53 CI checks · Track D: Compliance docs start

#### R-02-01 Create /outils/ hub page with 6 tool cards

Engineer A · 2h

**DoD:** /outils loads. 6 tool cards link to individual routes. Metadata present.

**Depends on:** R-01-05

#### R-02-02 Create 6 clinical tool routes under /outils/

Engineer A · 4h

**DoD:** /outils/posologie-pediatric, /outils/fonction-renale, /outils/grossesse-allaitement, /outils/ramadan, /outils/comparateur, /outils/armoire all load directly.

**Depends on:** R-02-01

#### R-02-03 Update Cmd-K palette (#11) to use router.push instead of setState

Engineer A · 2h

**DoD:** Cmd-K commands navigate to real routes. Single-key hotkeys (H, R, P, I, C, S, B, T) call router.push(). Selection state (selectedIndex) tracked correctly — Enter activates the highlighted command, not the first one.

**Depends on:** R-01-05

#### R-02-04 Create app/sitemap.ts (dynamic sitemap with all routes)

Engineer A · 2h

**DoD:** sitemap.xml lists all 11 static routes. Medication pages added in Phase 3. URL is https://dzpharm.dz/sitemap.xml (or vercel URL for now).

**Depends on:** R-02-02

#### C-02-01 Set up automated CI contrast checks via axe-core

Engineer B · 4h

**DoD:** CI pipeline runs axe-core on every PR that modifies globals.css. If contrast fails, CI blocks merge. Configured in GitHub Actions or Vercel CI.

**Depends on:** D-01-01

#### PM-02-01 Draft Terms of Service (updated for per-user accounts)

PM · 3h

**DoD:** ToS includes: platform is decision-support tool, user is responsible for clinical decisions, platform logs actions for regulatory compliance. Legal review scheduled.

**Depends on:** None

#### PM-02-02 Draft Privacy Policy (Law 18-07 compliant)

PM · 3h

**DoD:** Privacy policy includes: what data is collected (email, password hash, license number, audit logs), why (auth, accountability, compliance), retention (7 years for audit), user rights (access, rectification, erasure), DPA with Vercel/Neon.

**Depends on:** None

## Sprint 3 · Week 3 · Dynamic Medication Pages + Same-Day Three

Track A: #1 Routing Phase 3 (9,555 med pages) + #2 State same-day three · Track B: Storybook setup

#### R-03-01 Create app/medicament/[slug]/page.tsx with generateStaticParams + ISR

Engineer A · 4h

**DoD:** 9,555 medication pages pre-rendered (ISR, revalidate: 86400). /medicament/doliprane-500mg-comprime loads. generateMetadata produces unique title/description per medication. Custom 404 for unknown slugs.

**Depends on:** R-02-04

#### R-03-02 Create intercepting route for medication detail modal

Engineer A · 4h

**DoD:** app/repertoire/@modal/(.)medicament/[slug]/page.tsx renders modal version. Clicking a medication in /repertoire opens modal AND updates URL. Refresh on modal URL loads full page version. Both use same data-fetching.

**Depends on:** R-03-01

#### R-03-03 Update sitemap.ts to include all 9,555 medication pages

Engineer A · 1h

**DoD:** sitemap.xml lists 9,555 + 11 = 9,566 URLs. Each has lastModified, priority, changeFrequency.

**Depends on:** R-03-01

#### R-03-04 Create app/medicament/[slug]/error.tsx (custom branded 404)

Engineer A · 1h

**DoD:** Unknown slug shows DzPharm-branded 404: 'Médicament introuvable' + link to /repertoire. No raw Next.js error page.

**Depends on:** R-03-01

#### S-03-01 Create useInteractionListState hook (drug list in URL params)

Engineer A · 4h

**DoD:** Drug list reads from URL (?drugs=8421,1523). Adding/removing drugs updates URL via router.push. Patient context in URL (?ctx=...). Refresh preserves drug list. URL is shareable.

**Depends on:** R-01-03

#### S-03-02 Add copilotMessages + copilotMode slices to Zustand store (v2 → v3 migration)

Engineer A · 2h

**DoD:** Zustand store version bumped to 3. migrate() function handles v2 → v3 (adds copilotMessages=[], copilotMode='pro'). Copilot conversation persists across refresh. «Effacer la conversation» button clears store.

**Depends on:** None

#### S-03-03 Add useCrossTabSync hook (storage event listener)

Engineer A · 2h

**DoD:** Favorites added in Tab A appear in Tab B within 1 second. Theme toggle in Tab A syncs to Tab B. storage event → Zustand rehydrate().

**Depends on:** S-03-02

#### D-03-01 Initialize Storybook in the project

Designer · 4h

**DoD:** Storybook runs (npm run storybook). 'Colors' page shows all 12 semantic tokens as swatches. 'States' page shows 7-level state system. Basic structure ready for Phase 2 population.

**Depends on:** None

# Sprint 4 · Week 4 · Routing Complete + State Fixes + CI

Track A: #1 Phase 4 + #2 recently-viewed fix · Track B: Storybook population · Track D: Feature flag rollout begins

## R-04-01 Create 4 supporting pages (a-propos, contact, legal, sitemap)

Engineer A · 3h

**DoD:** /a-propos, /contact, /legal all load. /legal contains the ToS and Privacy Policy from PM-02-01/02. All 9,570 routes work.

**Depends on:** R-03-03

## S-04-01 Fix recentlyViewed cap (raise from 1 to 10, fix dedup bug)

Engineer A · 1h

**DoD:** Viewing 10 medications → recentlyViewed array has 10 entries. Dedup works (viewing same med twice doesn't duplicate). Cap enforced after dedup, not before.

**Depends on:** S-03-02

## S-04-02 Add audit logging infrastructure (preparation for #8)

Engineer A · 3h

**DoD:** logAction() helper created. Instrumented on medication view, Copilot query, interaction check (writes to console for now — will switch to Prisma after #8 Phase 4).

**Depends on:** None

## D-04-01 Populate Storybook: type scale (8 steps), weights (6), tracking (5)

Designer · 4h

**DoD:** Storybook 'Typography' page shows all 8 type scale steps with computed line-heights. 'Weights' page shows 6 weights. 'Tracking' page shows 5 levels. Each with sample text.

**Depends on:** D-03-01

## D-04-02 Populate Storybook: shadows (9 levels), palettes (20)

Designer · 4h

**DoD:** Storybook 'Shadows' page shows 9 shadow levels (6 standard + 3 themed glows). 'Palettes' page shows all 20 Nuancier palettes with names and descriptions.

**Depends on:** D-04-01

## PM-04-01 Begin feature flag rollout for Routing Phase 1-3

PM · 2h

**DoD:** Flag 'new-routing' enabled for 10% of users. Monitor error rate < 0.1%. Monitor 404 rate (should decrease). Daily check for 3 days.

**Depends on:** R-04-01, PM-01-01

# Sprint 5 · Week 5 · Auth Phase 1 + Copilot Bug Fixes

Track A: #8 Auth Phase 1 (NextAuth.js, httpOnly cookie) · Track B: #36 Copilot Phase 1 (5 bug fixes)

## A-05-01 Install NextAuth.js + create route handler with Credentials provider

Engineer A · 4h

**DoD:** npm install next-auth @auth/prisma-adapter. app/api/auth/[...nextauth]/route.ts created. Credentials provider validates code against process.env.DZPHARM\_ACCESS\_CODE. Session is JWT in httpOnly cookie.

**Depends on:** None

## A-05-02 Move shared code to process.env.DZPHARM\_ACCESS\_CODE

Engineer A · 30 min

**DoD:** Code is no longer in client-side code. process.env.DZPHARM\_ACCESS\_CODE set in Vercel env vars. Shared code still works as transitional login.

**Depends on:** A-05-01

## A-05-03 Replace sessionStorage with useSession() across the app

Engineer A · 4h

**DoD:** Header uses useSession() instead of sessionStorage.dzpharm\_session. Lock button calls signOut(). Devtools bypass closed (httpOnly cookie cannot be set via JS). Verified: document.cookie does not contain the session token.

**Depends on:** A-05-01

## A-05-04 Add rate limiting to /api/auth (Upstash Redis)

Engineer A · 4h

**DoD:** After 5 wrong attempts in 10 minutes, IP is rate-limited (429 response). Exponential backoff. Upstash Redis configured.

**Depends on:** A-05-01

## P-05-01 Fix Copilot version label (verify 3.6 vs 3.8, update both locations)

Engineer B · 30 min

**DoD:** Cmd-K label and Copilot header show the same version. Verified in both locations.

**Depends on:** None

## P-05-02 Wire suggested-prompt onClick handlers (populate + submit)

Engineer B · 1h

**DoD:** Clicking a suggested prompt calls setInput(prompt.text) + sendMessage(prompt.text). Prompt text appears in input. Query is sent immediately.

**Depends on:** None

## P-05-03 Upgrade system prompt with MODE\_PROMPTS (fix Mode patient)

Engineer B · 4h

**DoD:** buildSystemPrompt(mode, ctx) function created. Mode patient produces simplified output (brand names, no DCI, layperson language). Mode enfant still works. Mode pro unchanged. Verified: same query in 3 modes produces 3 different responses.

**Depends on:** None

## P-05-04 Persist Copilot mode to Zustand (fixes #37 gap)

Engineer B · 1h

**DoD:** Copilot mode stored in copilotMode slice. Survives refresh. Defaults to 'pro'.

**Depends on:** S-03-02



# Sprint 6 · Week 6 · Auth Phase 2 + Copilot Streaming

Track A: #8 Auth Phase 2 (per-user accounts, Prisma) · Track B: #36 Copilot Phase 2 (Vercel AI SDK streaming)

## A-06-01 Set up Prisma + PostgreSQL (Neon) + User model

Engineer A · 4h

**DoD:** prisma/schema.prisma created. User model (id, email, passwordHash, role, licenseNumber, licenseVerified, emailVerified, mfaEnabled, mfaSecret). Database connected (Neon free tier or Vercel Postgres). prisma db push successful.

**Depends on:** A-05-01

## A-06-02 Create registration page (email + password + role selection)

Engineer A · 4h

**DoD:** /register page: email, password, role dropdown (PHARMACIST/DOCTOR/PATIENT). For PHARMACIST/DOCTOR: license number field. Password hashed with bcrypt (12 rounds). User created with emailVerified=null.

**Depends on:** A-06-01

## A-06-03 Implement email verification (verification link sent on registration)

Engineer A · 4h

**DoD:** On registration, email sent with verification link (24h expiry). Clicking link sets emailVerified=now(). Login blocked until emailVerified is set.

**Depends on:** A-06-02

## A-06-04 Update login form to email + password (with shared-code fallback link)

Engineer A · 2h

**DoD:** Login form has email + password fields. 'Still have the shared code? Use it here' link below. Shared code still works as transitional login (deprecated).

**Depends on:** A-06-03

## P-06-01 Install Vercel AI SDK (ai + @ai-sdk/google)

Engineer B · 1h

**DoD:** npm install ai @ai-sdk/google. Packages installed. No breaking changes to existing code.

**Depends on:** None

## P-06-02 Refactor /api/copilot route to use streamText (streaming)

Engineer B · 4h

**DoD:** Route handler uses streamText() instead of generateContent(). Returns result.toDataStreamResponse(). onFinish callback logs to audit (console for now). maxDuration: 30.

**Depends on:** P-06-01, P-05-03

## P-06-03 Refactor Copilot component to use useChat hook

Engineer B · 4h

**DoD:** Component uses useChat({ api: '/api/copilot', body: { mode } }). Messages render progressively. Stop button appears during streaming. Stop() aborts fetch. Partial response preserved with '(interrompu)' label.

**Depends on:** P-06-02

P-06-04

### Add progressive markdown rendering (react-markdown + remark-gfm + react-katex)

Engineer B · 3h

**DoD:** Streaming tokens render as markdown in real-time. LaTeX deferred until complete ... block. Tables render when closing row arrives. No raw markdown visible during streaming.

**Depends on:** P-06-03

## Sprint 7 · Week 7 · Auth Phase 3 (MFA) + Copilot History

Track A: #8 Auth Phase 3 (TOTP MFA) · Track B: #36 Copilot Phase 3 (conversation history)

A-07-01

### Install otplib + create TOTP enrollment page

Engineer A · 4h

**DoD:** /settings/mfa page: generates TOTP secret, displays QR code (otpauth URL). User scans with Google Authenticator. User enters 6-digit code to verify. On success: mfaEnabled=true, mfaSecret stored (encrypted).

**Depends on:** A-06-01

A-07-02

### Add MFA step to login flow

Engineer A · 4h

**DoD:** After password verification, if mfaEnabled: prompt for 6-digit TOTP code. verifyTotp() checks against encrypted secret. On success: issue JWT. On failure: 'Code MFA incorrect' error. 3 attempts before logout.

**Depends on:** A-07-01

A-07-03

### Generate backup codes (10 single-use, 8-char hex)

Engineer A · 2h

**DoD:** On MFA enrollment: 10 backup codes generated. Displayed once (user must save). Each is single-use (consumed on use). If all used: admin reset required.

**Depends on:** A-07-01

A-07-04

### Add MFA recovery flow (admin reset via license verification)

Engineer A · 3h

**DoD:** Admin can reset MFA for a user (after identity verification: call pharmacy, verify license number). Reset clears mfaSecret, user must re-enroll. Logged to audit trail.

**Depends on:** A-07-03

P-07-01

### Wire Copilot to read/write from copilotMessages Zustand slice

Engineer B · 2h

**DoD:** useChat messages sync to Zustand copilotMessages on every change. On mount, messages restore from Zustand. «Effacer la conversation» button calls clearConversation(). Prune: last 30 messages.

**Depends on:** S-03-02, P-06-03

P-07-02

### Add mode-specific suggested prompts (12 total: 4 per mode)

Engineer B · 2h

**DoD:** Mode pro: current 4 prompts. Mode patient: «À quoi sert ce médicament?», «Comment le prendre?», «Quels effets secondaires?», «دواء السكر؟». Mode enfant: «Comment donner ce médicament à mon enfant?», «Quelle dose pour 22 kg?», «Est-ce dangereux?», «Quand consulter?». Prompts change when mode changes.

**Depends on:** P-05-02

# Sprint 8 · Week 8 · Auth Phase 4 (Audit Log) + Copilot

## Darija + Cost

Track A: #8 Auth Phase 4 (audit log, RBAC) · Track B: #36 Copilot Phase 4 (Darija + cost optimization)

### A-08-01 Create AuditLog Prisma model + logAction() helper

Engineer A · 3h

**DoD:** AuditLog model: id, userId, action (enum), target, metadata (Json), ipAddress, createdAt. logAction() helper writes to AuditLog. @@index on [userId, createdAt] and [action, createdAt].

**Depends on:** A-06-01

### A-08-02 Instrument all clinical actions with audit logging

Engineer A · 4h

**DoD:** Medication view → MEDICATION\_VIEWED. Copilot query → COPILOT\_QUERY (with mode, tokens). Interaction check → INTERACTION\_CHECKED (with drugIds). Chifa simulation → CHIFA\_SIMULATED. Shortage report → SHORTAGE\_REPORTED. All logged to userId.

**Depends on:** A-08-01, S-04-02

### A-08-03 Create admin audit dashboard at /admin/audit

Engineer A · 4h

**DoD:** Super-admins see all actions. Filter by action type, user, date range. Per-user action history. Read-only (no edit/delete). Anomaly detection placeholder (concurrent sessions flag).

**Depends on:** A-08-02

### A-08-04 Add RBAC middleware (route protection by role)

Engineer A · 3h

**DoD:** middleware.ts checks role on protected routes. /admin → ADMIN, SUPER\_ADMIN only. /interactions → PHARMACIST, DOCTOR, ADMIN, SUPER\_ADMIN. /copilote → all roles. Unauthorized → redirect to /unauthorized.

**Depends on:** A-06-01

### P-08-01 Add Darija few-shot examples to system prompt

Engineer B · 2h

**DoD:** 2-3 examples of Darija input → Darija output appended to system prompt. Tested: «دوا تاع السكر؟» produces Darija response (not MSA). Response includes Algerian brand names in Darija.

**Depends on:** P-05-03

### P-08-02 Add response caching (Redis, 24h TTL, keyed by query hash + mode)

Engineer B · 4h

**DoD:** Common queries (e.g., 'posologie paracétamol enfant 20 kg') served from cache. Cache hit: instant response, no API call, no cost. 15-25% estimated hit rate. Cache miss: normal API call + cache write.

**Depends on:** P-06-02

### P-08-03 Add model selection (Flash-Lite for simple queries, Flash for complex)

Engineer B · 2h

**DoD:** If query < 50 chars and doesn't contain 'posologie', 'interaction', 'calcul': use gemini-3.8-flash-lite. Else: gemini-3.8-flash. ~50% cost reduction on simple queries.

**Depends on:** P-06-02

**P-08-04** Add maxOutputTokens limit (1000 tokens ≈ 750 words)

Engineer B · 30 min

**DoD:** streamText called with maxTokens: 1000. Prevents runaway responses. Copilot's 4-section format is typically ~500-700 tokens.

**Depends on:** P-06-02

## Sprint 9 · Week 9 · State Phase 3 + Color Governance

Track A: #2 State Phase 3 (after #1 ships — URL state + intercepting routes) · Track B+C: #53 governance docs

**S-09-01** Migrate active view from setState to URL (closes Gap #1)

Engineer A · 2h

**DoD:** No more setActiveView() calls anywhere. All navigation uses next/link or router.push. Gap #1 fully closed (was dependent on #1 Phase 1 which shipped in Sprint 1-4).

**Depends on:** R-04-01 (already shipped)

**S-09-02** Migrate medication detail from modal useState to intercepting route (closes Gap #2)

Engineer A · 2h

**DoD:** No more selectedMed useState. Medication detail opens via Link to /medicament/[slug]. Modal intercepts. Gap #2 fully closed.

**Depends on:** R-03-02 (already shipped)

**S-09-03** Add idle timeout (15 min inactivity → lock)

Engineer A · 2h

**DoD:** useIdleTimer hook: 15 minutes of no mousemove/keydown/click → signOut(). Pharmacist must re-authenticate. Closes #9 Session Management gap.

**Depends on:** A-05-03

**S-09-04** Add already-authenticated redirect (/login → / if session exists)

Engineer A · 1h

**DoD:** Visiting /login while authenticated redirects to /. No login form shown to already-logged-in users.

**Depends on:** A-05-03

**C-09-01** Write palette contribution guide

Engineer B · 2h

**DoD:** Guide documents: how to add a new Nuancier palette (define object, verify contrast, verify color-blindness, add to store, add to dialog). 5-step checklist.

**Depends on:** D-04-02

**C-09-02** Create token reference page (/design-system/tokens, admin-only)

Engineer B · 3h

**DoD:** Page renders all ~400 tokens as swatches. Shows name, value, contrast ratio. Admin-only route. Serves as quick reference for developers.

**Depends on:** D-04-02

## Sprint 10 · Week 10 · State Phase 4 + Color Complete + Text-Only Mode

Track A: #2 State Phase 4 (server-side after #8) · Track B: #53 complete + text-only palette

**S-10-01** Create Prisma models for server-side state (Favorite, CopilotConversation, CopilotMessage, RecentlyViewed) Engineer A · 3h

**DoD:** Models created with userId foreign key. CopilotConversation (id, userId, title, mode, messages[]). CopilotMessage (id, conversationId, role, content, mode, tokenCount). Favorite (userId, medId). RecentlyViewed (userId, medId, viewedAt, viewCount).

**Depends on:** A-06-01

**S-10-02** Implement server-side sync (optimistic updates + debounce 300ms) Engineer A · 4h

**DoD:** On state change: Zustand updates immediately (UI reflects change). Debounced server sync (300ms). On login: fetch server state, merge with local. If sync fails: keep local, retry.

**Depends on:** S-10-01

**S-10-03** Add Copilot thread list (sidebar of past conversations) Engineer A · 4h

**DoD:** Sidebar shows past conversations (title = first message, truncated). Click to resume. Conversations sync across devices (server-side). Search by keyword (Phase 3).

**Depends on:** S-10-01, S-10-02

**C-10-01** Add text-only (monochrome) Nuancier palette Engineer B · 1 day

**DoD:** 21st palette: strips all color from status indicators. Uses text labels + icons only. For users with severe color vision deficiencies. Tested: all information conveyable without color.

**Depends on:** C-09-01

**D-10-01** Complete Storybook: all components documented Designer · 1 day

**DoD:** Every DzPharm component has a Storybook entry. Props documented. States documented (hover, focus, disabled, loading). Storybook deployed (Vercel or Chromatic).

**Depends on:** D-04-02

## Sprint 11 · Week 11 · Integration Testing

Both tracks: end-to-end testing of all 6 pillars together. Storybook completion. Governance docs.

**X-11-01** End-to-end test: full user journey (register → login → MFA → browse → Copilot → favorites → logout) Engineer A · 4h

**DoD:** Test script covers: register new pharmacist, verify email, set up MFA, login with MFA, browse Répertoire, search doliprane, open medication detail (intercepting route), ask Copilot a query (streaming), favorite a medication, refresh (state persists), lock session, logout. All steps pass.

**Depends on:** All pillars

### X-11-02 End-to-end test: SEO crawl (Google Search Console simulation)

Engineer A · 2h

**DoD:** Fetch /sitemap.xml — contains 9,570 URLs. Fetch /medicament/doliprane-500mg-comprime — returns 200 with unique title/meta. Fetch /repertoire?q=doliprane — returns 200 with search results. JSON-LD Drug schema present on medication pages.

**Depends on:** R-04-01

### X-11-03 End-to-end test: a11y (keyboard-only + screen reader)

Engineer A · 4h

**DoD:** Tab through every page. Skip-link works. All modals trap focus. All icon-only buttons have aria-labels. role=alert announces errors. role=status announces Copilot response. prefers-reduced-motion disables animations.

**Depends on:** All pillars

### X-11-04 End-to-end test: security (OWASP Top 10 checklist)

Engineer A · 4h

**DoD:** Devtools bypass: blocked (httpOnly cookie). Brute force: rate-limited after 5 attempts. MFA bypass: blocked. Session fixation: not vulnerable. CSRF: NextAuth.js built-in protection. XSS: React escapes by default.

**Depends on:** A-08-04

### D-11-01 Final Storybook review + publish

Designer · 4h

**DoD:** Storybook reviewed by team. All tokens, components, states documented. Published to Vercel/Chromatic. URL shared with team.

**Depends on:** D-10-01

## Sprint 12 · Week 12 · Feature Flag Rollout + Polish

Track D: gradual rollout 10% → 50% → 100%. Both tracks: final polish, documentation.

### PM-12-01 Enable all feature flags for 10% of users

PM · 1h

**DoD:** Flags 'new-routing', 'new-auth', 'new-copilot' enabled for 10%. Monitor error rate < 0.1%. Monitor user feedback. Daily check for 3 days.

**Depends on:** X-11-01 through X-11-04

### PM-12-02 Enable for 50% of users

PM · 1h

**DoD:** Flags enabled for 50%. Monitor error rate < 0.05%. Monitor Copilot cost (should be ~\$2-3/day at 50%). Daily check for 3 days.

**Depends on:** PM-12-01 (3 days stable)

### PM-12-03 Enable for 100% of users

PM · 1h

**DoD:** Flags enabled for 100%. Full rollout. Monitor for 1 week.

**Depends on:** PM-12-02 (3 days stable)

### X-12-01 Remove deprecated setState code (cleanup)

Engineer A · 2h

**DoD:** All setActiveView() calls removed. Old useState navigation code deleted. Codebase is clean — no legacy patterns remain.

**Depends on:** PM-12-03

**X-12-02 Remove shared-code fallback (deprecate transitional auth)**

Engineer A · 1h

**DoD:** Shared code login removed. All users must use email + password + MFA. Transition complete.

**Depends on:** PM-12-03

**D-12-01 Final visual QA across all 9,570 routes**

Designer · 4h

**DoD:** Spot-check 20 medication pages, 4 primary routes, 6 tool routes, login, register, admin dashboard. All render correctly in dark + light theme. No layout regressions.

**Depends on:** PM-12-03

## Sprint 13 · Week 13 · Full Rollout + Monitoring

Remove feature flags. Deploy to production. Monitor for 1 week. Platform grade: A+.

**PM-13-01 Remove all feature flags from code**

PM · 2h

**DoD:** Feature flag checks removed from code. Flags removed from Vercel Edge Config. Code is clean — no conditional routing/auth/copilot logic.

**Depends on:** PM-12-03 (1 week stable at 100%)

**PM-13-02 Submit sitemap to Google Search Console**

PM · 30 min

**DoD:** sitemap.xml submitted to Google Search Console. Request indexing for 9,570 pages. Monitor indexing progress over following weeks.

**Depends on:** R-04-01

**PM-13-03 Submit to Bing Webmaster Tools**

PM · 30 min

**DoD:** sitemap.xml submitted to Bing Webmaster Tools. Monitor indexing.

**Depends on:** R-04-01

**X-13-01 Monitor: error rate, Copilot cost, user registrations, organic traffic**

Engineer A · Ongoing

**DoD:** Daily monitoring for 1 week: Vercel error rate < 0.05%. Copilot cost < \$5/day. User registrations trending up. Organic traffic increasing (Google indexing begins).

**Depends on:** PM-13-01

**X-13-02 Final grade assessment: verify all 6 pillars are A+**

Engineer A + B · 4h

**DoD:** Re-run the audit: 68-feature grade assessment. Verify: #1 Routing A+, #8 Auth A+, #2 State A+, #36 Copilot A+, #53 Color A+, #54 Typography A+. Overall platform grade: A+.

**Depends on:** X-13-01

# Ticket Summary & Risk Register

This appendix provides a quick-reference summary of all tickets, their status, and the risk register for the transformation.

## Ticket Count by Pillar

| PILLAR                        | TICKETS | TOTAL EFFORT | SPRINTS            |
|-------------------------------|---------|--------------|--------------------|
| #1 Routing (R-XX-YY)          | 12      | ~44h         | Sprint 1-4         |
| #8 Auth (A-XX-YY)             | 14      | ~50h         | Sprint 5-8         |
| #2 State (S-XX-YY)            | 10      | ~30h         | Sprint 3, 4, 9, 10 |
| #36 Copilot (P-XX-YY)         | 14      | ~38h         | Sprint 5-8         |
| #53 Color (C-XX-YY + D-XX-YY) | 8       | ~32h         | Sprint 1-4, 9-10   |
| #54 Typography (T-XX-YY)      | 1       | 30 min       | Sprint 0           |
| Independent (X-XX-YY)         | 8       | ~25h         | Sprint 0, 11-13    |
| PM (PM-XX-YY)                 | 10      | ~22h         | Sprint 1-4, 12-13  |
| Total                         | ~77     | ~241h        | 13 sprints         |



# Risk Register

| # | RISK                                     | PROBABILITY | IMPACT | MITIGATION                                                     | TICKET                  |
|---|------------------------------------------|-------------|--------|----------------------------------------------------------------|-------------------------|
| 1 | Routing migration breaks setState nav    | Medium      | High   | Feature flag, keep old code as fallback                        | PM-01-01, PM-12-01      |
| 2 | NextAuth migration breaks sessions       | Medium      | High   | Phase 1 keeps shared code, gradual transition                  | A-05-01 through A-06-04 |
| 3 | Gemini API cost exceeds budget           | Low         | Medium | Cost optimizations: pruning, caching, model selection          | P-08-02 through P-08-04 |
| 4 | 9,555 pages cause build timeout          | Medium      | Medium | Use ISR (revalidate: 86400), not SSG                           | R-03-01                 |
| 5 | Noto Sans Arabic increases payload       | Low         | Low    | Variable font, SW cache, font-display: swap                    | T-00-01                 |
| 6 | License API unavailable                  | High        | Medium | Admin manual review fallback                                   | A-06-02                 |
| 7 | Team capacity (2 engineers, 13 weeks)    | Medium      | High   | Day 1 quick wins ship regardless, parallel tracks              | All sprints             |
| 8 | Cultural resistance to per-user accounts | Medium      | Medium | Shared code fallback, gradual transition, communicate benefits | A-06-04                 |

## Definition of Done — Platform Level

THE PLATFORM IS DONE WHEN:

- 1. All 9,570 routes return 200 (no 404s on valid routes).
- 2. Users can register, verify email, set up MFA, and log in with email + password + TOTP.
- 3. Every clinical action (medication view, Copilot query, interaction check) is logged to userId with 7-year retention.
- 4. Drug list survives refresh (URL params). Copilot conversation survives refresh (Zustand). Favorites sync across tabs.
- 5. Copilot streams responses (~1s first token). Mode patient produces simplified output. Darija queries get Darija responses.
- 6. Inter + Noto Sans Arabic loaded (document.fonts.size === 3). Brand identity stable across devices.
- 7. Storybook documents all tokens, components, and states. CI prevents contrast regression.
- 8. Sitemap submitted to Google. Feature flags removed. Error rate < 0.05%.
- 9. Overall platform grade: A+. Zero D-grade features. Zero P0 issues.

## Source Documents

| DOCUMENT                                   | PAGES      | CONTENT                                                        |
|--------------------------------------------|------------|----------------------------------------------------------------|
| Platform Transformation Blueprint          | 21         | Executive summary, timeline, ROI, team assignments             |
| #1 Routing Architecture (Technical Design) | 26         | 4-phase migration, intercepting routes, ISR, sitemap           |
| #8 Auth Model (Technical Design)           | 28         | Threat model, NextAuth.js, MFA, audit log, compliance          |
| #2 State Management (Technical Design)     | 27         | 6 gaps, Zustand v3, same-day three, cross-tab sync             |
| #36 Copilot Core (Technical Design)        | 27         | Vercel AI SDK, system prompt, Darija, cost optimization        |
| #53 Color System (Technical Design)        | 32         | 400 tokens, 7-state system, 20 palettes, Storybook             |
| #54 Typography (Technical Design)          | 29         | 8-step scale, calc() line-heights, next/font, Arabic           |
| Executive Presentation (PPT)               | 15 slides  | Boardroom-ready summary of the transformation                  |
| <b>This Sprint Planning Document</b>       | <b>~40</b> | <b>77 tickets, 13 sprints, risk register, DoD</b>              |
| <b>Total</b>                               |            | <b>~259 pages of implementation-ready planning + 15 slides</b> |

### START SPRINT 0 TODAY

Ship the 3 Day 1 quick wins (T-00-01, X-00-02, X-00-03) in 1 hour. Then begin Sprint 1 on Monday. The full 13-week plan is ~77 tickets, ~241 hours of engineering, and transforms the platform from B- to A+. Every ticket has a definition-of-done, dependencies, and risk flag. Copy the ticket IDs into Jira and start sprinting.

# 77

77 tickets across 13 sprints. Every ticket has an ID, assignee, effort, dependencies, definition-of-done, and risk flag. Jira-ready. ~241 hours of engineering transforms the platform from B- to A+.

**13**

SPRINTS

**77**

TICKETS

**~241h**

ENGINEERING HOURS

**A+**

TARGET GRADE